

Problem Set-6

Problem 1. One way of detecting errors is to transmit data as a block of n rows of k bits per row and adding parity bits to each row and each column. The lower-right corner is a parity bit that checks its row and its column. Will this scheme detect all single errors? Double errors? Triple errors?

Problem 2. A bit stream 10011101 is transmitted using the standard CRC method described in the text. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted. Suppose the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.

Problem 3. An 8-bit byte with binary value 10101111 is to be encoded using an even-parity Hamming code. What is the binary value after encoding?

Problem 4. Consider the use of 1000-bit frames on a 1-Mbps satellite channel with a 270-ms delay. What is the maximum link utilization for

- a. Stop-and-wait flow control?
- b. Continuous flow control with a window size of 7?
- c. Continuous flow control with a window size of 127?
- d. Continuous flow control with a window size of 255?